

LOXOF TABLETS

(LEVOFLOXACIN TABLETS 500 mg)

COMPOSITION

Each film-coated tablet contains:

Levofloxacin hemihydrates equivalent to Levofloxacin..... 500 mg

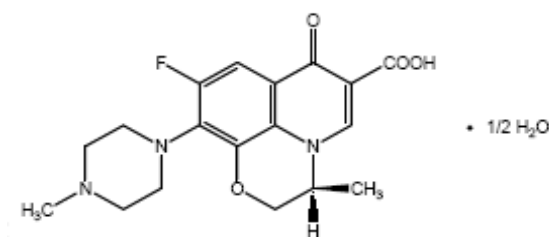
Inactive ingredients: Crospovidone, Microcrystalline cellulose, Opadry 03B52874, Polysorbate 80, Magnesium Stearate, Hypromellose and Purified water.

PRODUCT DESCRIPTION

Light peach coloured, film coated, biconvex caplets debossed with '500' on one side

DESCRIPTION

LOXOF TABLETS contains levofloxacin. Levofloxacin is a synthetic broad spectrum antibacterial agent. Chemically, levofloxacin, a chiral fluorinated carboxyquinolone, is the pure (-)-(S)-enantiomer of the racemic drug substance ofloxacin. The chemical name for levofloxacin hemihydrate is (-)-(S)-9-fluoro-2,3-dihydro-3-methyl-10-(4-methyl-1-piperazinyl)-7-oxo-7H-pyrido(1,2,3-de)-1,4-benzoxazine-6-carboxylic acid hemihydrate. The empirical formula for levofloxacin hemihydrate is $C_{18}H_{20}FN_3O_4 \cdot \frac{1}{2}H_2O$ and its molecular weight is 370.38. Its structural formula is as given below:



Levofloxacin Hemihydrate

PHARMACOLOGY

Mechanism of action

Levofloxacin is a member of the fluoroquinolone class of antibacterial agents. Levofloxacin is the L-isomer of the racemate, ofloxacin, a quinolone antimicrobial agent. The antibacterial activity of ofloxacin resides primarily in the L-isomer. The mechanism of action of levofloxacin and other fluoroquinolone antimicrobials involves inhibition of bacterial topoisomerase IV and DNA gyrase (both of which are type II topoisomerases), enzymes required for DNA replication, transcription, repair and recombination.

PK/PD Relationship

The degree of the bactericidal activity of levofloxacin depends on the ratio of the maximum concentration in serum (C_{max}) or the area under the curve (AUC) and the minimal inhibitory concentration (MIC).

Mechanism of resistance

Resistance to levofloxacin is acquired through a stepwise process by target site mutations in both type II topoisomerases, DNA gyrase and topoisomerase IV. Other resistance mechanisms such as permeation barriers (common in *Pseudomonas aeruginosa*) and efflux mechanisms may also affect susceptibility to levofloxacin.

Cross-resistance between levofloxacin and other fluoroquinolones is observed. Due to the mechanism of action, there is generally no cross-resistance between levofloxacin and other classes of antibacterial agents.

The prevalence of resistance may vary geographically and with time for selected species and local information on resistance is desirable, particularly when treating severe infections. As necessary, expert advice should be sought when the local prevalence of resistance is such that the utility of the agent in at least some types of infections is questionable.

Commonly susceptible species

Aerobic Gram-positive bacteria

Bacillus anthracis
Staphylococcus aureus methicillin-susceptible
Staphylococcus saprophyticus
Streptococci, group C and G
Streptococcus agalactiae
Streptococcus pneumoniae
Streptococcus pyogenes

Aerobic Gram-negative bacteria

Eikenella corrodens
Haemophilus influenzae

Haemophilus para-influenzae
Klebsiella oxytoca
Moraxella catarrhalis
Pasteurella multocida
Proteus vulgaris
Providencia rettgeri

Anaerobic bacteria

Peptostreptococcus

Other

Chlamydophila pneumoniae
Chlamydophila psittaci
Chlamydia trachomatis
Legionella pneumophila
Mycoplasma pneumoniae
Mycoplasma hominis
Ureaplasma urealyticum

Species for which acquired resistance may be a problem

Aerobic Gram-positive bacteria

Enterococcus faecalis
Staphylococcus aureus methicillin-resistant[#]
Staphylococcus coagulase spp

Aerobic Gram- negative bacteria

Acinetobacter baumannii
Citrobacter freundii^{*}
Enterobacter aerogenes
Enterobacter cloacae
Escherichia coli
Klebsiella pneumoniae
Morganella morganii
Proteus mirabilis
Providencia stuartii
Pseudomonas aeruginosa
Serratia marcescens

Anaerobic bacteria

Bacteroides fragilis

Inherently Resistant Strains

Aerobic Gram-positive bacteria

Enterococcus faecium

Methicillin-resistant *S. aureus* are very likely to possess co-resistance to fluoroquinolones, including levofloxacin.

Pharmacokinetic Properties

Absorption

Levofloxacin is rapidly and essentially completely absorbed after oral administration. Peak plasma concentrations are usually attained one to two hours after oral dosing. The absolute bioavailability of a 500 mg tablet and a 750 mg tablet of levofloxacin are both approximately 99%, demonstrating complete oral absorption of levofloxacin. Following a single intravenous dose of levofloxacin to healthy volunteers, the mean $\pm$ SD peak plasma concentration attained was 6.2 ± 1.0 mcg/mL after a 500 mg dose infused over 60 minutes and 11.5 ± 4.0 mcg/mL after a 750 mg dose infused over 90 minutes.

Levofloxacin pharmacokinetics is linear and predictable after single and multiple oral/or i.v. dosing regimens. Steady-state conditions are reached within 48 hours following a 500 mg or 750 mg once-daily dosage regimen. The mean $\pm$ SD peak and trough plasma concentrations attained following multiple once-daily oral dosage regimens were approximately 5.7 ± 1.4 and 0.5 ± 0.2 μ g/mL after the 500 mg doses, and 8.6 ± 1.9 and 1.1 ± 0.4 μ g/mL after the 750 mg doses, respectively. The mean $\pm$ SD peak and trough plasma concentrations attained following multiple once-daily i.v. regimens were approximately 6.4 ± 0.8 and 0.6 ± 0.2 μ g/mL after the 500 mg doses, and 12.1 ± 4.1 and 1.3 ± 0.71 μ g/mL after the 750 mg doses, respectively.

Oral administration of a 500 mg levofloxacin tablet with food slightly prolongs the time to peak concentration by approximately 1 hour and decreases the peak concentration by approximately 14% following tablet and approximately 25% following oral solution administration. Therefore, levofloxacin tablets can be administered without regard to food.

The plasma concentration profile of levofloxacin after i.v. administration is similar and comparable in extent of exposure (AUC) to that observed for levofloxacin tablets when equal doses (mg/mg) are administered. Therefore, the oral and i.v. routes of administration can be considered interchangeable.

Distribution

The mean volume of distribution of levofloxacin generally ranges from 74 to 112 L after single and multiple 500 mg or 750 mg doses, indicating widespread distribution into body tissues. Levofloxacin reaches its peak levels in skin tissues and in blister fluid at approximately 3 hours after dosing. The skin tissue biopsy to plasma AUC ratio is approximately 2 and the blister fluid to plasma AUC ratio is approximately 1 following multiple once-daily oral administrations of 750 mg and 500 mg levofloxacin, respectively. Levofloxacin also penetrates well into lung tissues. Lung tissue concentrations were generally 2 to 5 fold higher than plasma concentrations and ranged from approximately 2.4 to 11.3 µg/g over a 24-hour period after a single 500 mg oral dose.

In vitro, over a clinically relevant range (1 to 10 µg/mL) of serum/plasma levofloxacin concentrations, levofloxacin is approximately 24 to 38% bound to serum proteins across all reported species studied. Levofloxacin is mainly bound to serum albumin in humans. Levofloxacin binding to serum proteins is independent of the drug concentration.

Metabolism

Levofloxacin is stereochemically stable in plasma and urine, and does not invert metabolically to its enantiomer, D-ofloxacin. Levofloxacin undergoes limited metabolism in humans and is primarily excreted as unchanged drug in the urine. Following oral administration, approximately 87% of an administered dose was recovered as unchanged drug in urine within 48 hours, whereas less than 4% of the dose was recovered in faeces in 72 hours. Less than 5% of an administered dose was recovered in the urine as the desmethyl and N-oxide metabolites, the only metabolites identified in humans. These metabolites have little relevant pharmacological activity.

Excretion

Levofloxacin is excreted largely as unchanged drug in the urine. The mean terminal plasma elimination half-life of levofloxacin ranges from approximately 6 to 8 hours following single or multiple doses of levofloxacin given orally or intravenously. The mean apparent total body clearance and renal clearance range from approximately 144 to 226 mL/min and 96 to 142 mL/min, respectively. Renal clearance in excess of the glomerular filtration rate suggests that tubular secretion of levofloxacin occurs in addition to its glomerular filtration. Concomitant administration of either cimetidine or probenecid results in approximately 24% and 35% reduction in the levofloxacin renal clearance, respectively, indicating that secretion of levofloxacin occurs in the renal proximal tubule. No levofloxacin crystals were found in any of the urine samples freshly collected from subjects receiving levofloxacin.

Special populations

Geriatric: There are no significant differences in levofloxacin pharmacokinetics between young and elderly subjects when the subjects' differences in creatinine clearance are taken into consideration. Following a 500 mg oral dose of levofloxacin to healthy elderly subjects (66 to 80 years of age), the mean terminal plasma elimination half-life of levofloxacin was about 7.6 hours, as compared to approximately 6 hours in younger adults. The difference was attributable to the variation in renal function status of the subjects and was not believed to be clinically significant. Drug absorption appears to be unaffected by age. Levofloxacin dose adjustment based on age alone is not necessary.

Gender: There are no significant differences in levofloxacin pharmacokinetics between male and female subjects when subjects' differences in creatinine clearance are taken into consideration. Following a 500 mg oral dose of levofloxacin to healthy male subjects, the mean terminal plasma elimination half-life of levofloxacin was about 7.5 hours, as compared to approximately 6.1 hours in female subjects. This difference was attributable to the variation in renal function status of the male and female subjects and was not believed to be clinically significant. Drug absorption appears to be unaffected by the gender of the subjects. Dose adjustment based on gender alone is not necessary.

Race: The apparent total body clearance and apparent volume of distribution of levofloxacin is not affected by the race of the subjects.

Renal insufficiency: Clearance of levofloxacin is substantially reduced and plasma elimination half-life is substantially prolonged in patients with impaired renal function (creatinine clearance <50 mL/min), requiring dosage adjustment in such patients to avoid accumulation. Neither haemodialysis nor continuous ambulatory peritoneal dialysis (CAPD) is effective in removal of levofloxacin from the body, indicating that supplemental doses of levofloxacin are not required following haemodialysis or CAPD.

Hepatic insufficiency: Pharmacokinetic studies in hepatically impaired patients have not been reported. Due to the limited extent of levofloxacin metabolism, the pharmacokinetics of levofloxacin are not expected to be affected by hepatic impairment.

Bacterial infection: The pharmacokinetics of levofloxacin in patients with serious community-acquired bacterial infections are comparable to those observed in healthy subjects.

Drug-drug interactions: The potential for pharmacokinetic drug interactions between levofloxacin and antacids, warfarin, theophylline, cyclosporine, digoxin, probenecid, and cimetidine has been reported (see **DRUG INTERACTIONS**).

INDICATIONS

Consideration should be given to official guidance on the appropriate use of antibacterial agents

Treatment of adults (more than or equal to 18 years) with mild, moderate and severe infections caused by susceptible strains of the designated micro-organisms in the conditions listed as follows. Levofloxacin injection is indicated when i.v. administration offers a route of administration advantageous to the patients (e.g. patients can not tolerate an oral dosage form), (see **DOSAGE AND ADMINISTRATION**)

- Acute maxillary sinusitis due to *Streptococcus pneumoniae*, *Haemophilus influenza* or *Moraxella catarrhalis*.
- Acute bacterial exacerbation of chronic bronchitis due to *Staphylococcus aureus*, *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Haemophilus parainfluenzae* or *Moraxella catarrhalis* *.
- Community acquired pneumonia due to *Staphylococcus aureus*, *Streptococcus pneumoniae*, *Legionella pneumophila* or *Mycoplasma pneumoniae*.
- Uncomplicated skin and skin structure infections (mild to moderate) including abscess, cellulitis, furuncles, impetigo, pyoderma, wound infections due to *Staphylococcus aureus* or *Streptococcus pyogenes*.
- Complicated urinary tract infections (mild to moderate) due to *Enterococcus faecalis*, *Enterobacter cloacae*, *Escherichia coli*, *Klebsiella pneumoniae*, *Proteus mirabilis* or *Pseudomonas aeruginosa* *.
- Acute pyelonephritis (mild to moderate) caused by *Escherichia coli*.

Appropriate culture and susceptibility tests should be performed before treatment in order to isolate and identify organisms causing the infection and to determine their susceptibility to levofloxacin. Therapy with levofloxacin may be initiated before results of these tests are known; once results become available, appropriate therapy should be selected.

As with other drugs in this class, some isolates of *Pseudomonas aeruginosa* may develop resistance fairly rapidly during treatment with levofloxacin. Culture and susceptibility testing performed periodically during therapy will provide information about the continued susceptibility of the pathogens to the antimicrobial agent and also the possible emergence of bacterial resistance.

* **LOXOF TABLETS** should be only used:

- When *Pseudomonas* is considered and the patients is allergic to antipseudomonal penicillins/ cephalosporins
- For the resistant organism with no other alternative antibiotics available.

DOSAGE AND ADMINISTRATION

LOXOF TABLETS tablets are available in the strength of 500 mg only and may not be suitable for all dosage recommendations given below. Therefore, other suitable available strengths and/or dosage forms of levofloxacin should be used in such cases.

The usual dose of levofloxacin is 250 mg or 500 mg administered orally every 24 hours, as indicated by infection and described in following table. These recommendations apply to patients with normal renal function (i.e. creatinine clearance > 80 mL/min).

For patients with altered renal function, see table below, oral doses should be administered at least two hours before or two hours after antacids containing magnesium, aluminium, as well as sucralfate, metal cation such as iron and multivitamin preparation with zinc and didenosine, chewable/buffered tablets or the paediatric powder for oral solution.

Table: Patient with Normal Renal Function

Infection*	Unit dose (mg)	Frequency	Duration* (Days)	Daily Dose (mg)
Acute Bacterial Exacerbation of Chronic Bronchitis	500	every 24 hours	7	500
Community Acquired Pneumonia	500	every 24 hours	7-14	500
Acute maxillary sinusitis	500	every 24 hours	10-14	500
Uncomplicated skin and skin structure infection	500	every 24 hours	7-10	500
Complicated UTI	250	every 24 hours	10	250
Acute pyelonephritis	250	every 24 hours	10	250

*Due to designated pathogens (see **INDICATIONS**)

Sequential therapy (IV or Oral) may be instituted at the discretion of the physician.

Table: Patient with Impaired Renal Function

Renal Status	Initial Dose (mg)	Subsequent Dose (mg)
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Acute Bacterial Exacerbation of Chronic Bronchitis, Community Acquired Pneumonia, Acute maxillary sinusitis, Uncomplicated skin and skin structure infection.		
Cl _{CR} 50-80 mL/min	No dosage adjustment	
Cl _{CR} 20-49mL/min	500 mg	250 mg every 24 hours
Cl _{CR} 10-19mL/min	500 mg	250 mg every 48 hours
Haemodialysis	500 mg	250 mg every 48 hours
CAPD	500 mg	250 mg every 48 hours
Complicated UTI, Acute pyelonephritis		
Cl _{CR} ≥ 20 mL/min	No dosage adjustment required	
Cl _{CR} 10-19mL/min	250 mg	250 mg every 48 hours

Cl_{CR}- Creatinine clearance; CAPD- chronic ambulatory peritoneal dialysis.

When only the serum creatinine is known, the following formula may be used to calculate creatinine clearance:

Males:

$$\text{Creatinine clearance (mL/min)} = \frac{\text{Weight (kg)} \times (140 - \text{age})}{72 \times \text{serum creatinine (mg/dL)}}$$

Females:

0.85 X value calculated using formula for males.

The serum creatinine should represent a steady state of renal function.

CONTRAINDICATIONS

LOXOF TABLETS is contraindicated:

- in persons with a known hypersensitivity to levofloxacin, quinolone antimicrobial agents, or to any component/excipient of this product.

WARNINGS AND PRECAUTIONS

BOXED WARNING

Fluoroquinolones, including levofloxacin, are associated with an increased risk of tendinitis and tendon rupture in all ages. This risk is further increased in older patients usually over 60 years of age, in patients taking corticosteroid drugs, and in patients with kidney, heart or lung transplants.

Fluoroquinolones, including levofloxacin, may exacerbate muscle weakness in persons with myasthenia gravis. Avoid levofloxacin in patients with a known history of myasthenia gravis.

The use of levofloxacin should be avoided in patients who have experienced serious reactions in the past when using fluoroquinolones containing products (see **ADVERSE REACTIONS**). Treatment of these patients with levofloxacin should be initiated in the absence of alternative treatment options and after careful benefit/risk assessment.

Tendinitis and tendon rupture

Tendinitis and tendon rupture (especially but not limited to Achilles tendon), sometimes bilateral, may occur as early as within 48 hours of starting treatment with fluoroquinolones and have been reported to occur even up to several months after discontinuation of treatment. The risk of tendinitis and tendon rupture is increased in older patients (above 60 years of age) with renal impairment, patients with solid organ transplants and those treated concurrently with corticosteroids*. Therefore, concomitant use of corticosteroids should be avoided.

At the first sign of tendinitis (e.g. painful swelling, inflammation) the treatment with levofloxacin should be discontinued and alternative treatment should be considered. The affected limb(s) should be appropriately treated (e.g. immobilisation). Corticosteroids should not be used if signs of tendinopathy occur.

*(for systemically administered levofloxacin-containing products, the listing of risk factors should additionally include: in patients receiving daily doses of 1000 mg levofloxacin.)

Exacerbation of myasthenia gravis

Fluoroquinolones, including levofloxacin, have neuromuscular blocking activity and may exacerbate muscle weakness in persons with myasthenia gravis. Postmarketing serious adverse events, including deaths and requirement for ventilatory support, have been associated with fluoroquinolone use in persons with myasthenia gravis. Avoid levofloxacin in patients with a known history of myasthenia gravis (see **ADVERSE REACTIONS**).

Serious and Sometimes Fatal Reactions

Other serious and sometimes fatal events, some due to hypersensitivity, and some due to uncertain etiology, have been reported rarely in patients receiving therapy with fluoroquinolones, including levofloxacin. These events may be severe and generally occur following the administration of multiple doses. Clinical manifestations may include one or more of the following:

- fever, rash, or severe dermatologic reactions (e.g., toxic epidermal necrolysis, Stevens-Johnson Syndrome);
- vasculitis; arthralgia; myalgia; serum sickness;
- allergic pneumonitis;
- interstitial nephritis; acute renal insufficiency or failure;
- hepatitis; jaundice; acute hepatic necrosis or failure;
- anaemia, including haemolytic and aplastic; thrombocytopenia, including thrombotic thrombocytopenic purpura; leukopenia; agranulocytosis; pancytopenia; and/or other hematologic abnormalities.

The drug should be discontinued immediately at the first appearance of skin rash, jaundice, or any other sign of hypersensitivity and supportive measures instituted.

Blood Glucose Disturbances

As with other fluoroquinolones, disturbances of blood glucose, including symptomatic hyper- and hypoglycemia, have been reported with levofloxacin, usually in diabetic patients receiving concomitant treatment with an oral hypoglycemic agent (e.g., glyburide) or with insulin. In these patients, careful monitoring of blood glucose is recommended. If a hypoglycemic reaction occurs in a patient being treated with levofloxacin, levofloxacin should be discontinued and appropriate therapy should be initiated immediately.

Central Nervous System Effects

Convulsions, toxic psychoses, increased intracranial pressure (including pseudotumor cerebri) have been reported in patients receiving fluoroquinolones, including levofloxacin. Fluoroquinolones may also cause central nervous system stimulation which may lead to tremors, restlessness, anxiety, lightheadedness, confusion, hallucinations, paranoia, depression, nightmares, insomnia, and, rarely, suicidal thoughts or acts. These reactions may occur following the first dose. If these reactions occur in patients receiving levofloxacin, the drug should be discontinued and appropriate measures instituted. As with other fluoroquinolones, levofloxacin should be used with caution in patients with a known or suspected central nervous system (CNS) disorder that may predispose them to seizures or lower the seizure threshold (e.g., severe cerebral arteriosclerosis, epilepsy) or in the presence of other risk factors that may predispose them to seizures or lower the seizure threshold (e.g., certain drug therapy, renal dysfunction).

Clostridium difficile-associated disease

Diarrhoea, particularly if severe, persistent and/or bloody, during or after treatment with levofloxacin tablets (including several weeks after treatment),, may be symptomatic of *Clostridium difficile*-associated disease (CDAD). CDAD may range in severity from mild to life threatening, the most severe form of which is pseudomembranous colitis. It is therefore important to consider this diagnosis in patients who develop serious diarrhoea during or after treatment with levofloxacin. If CDAD is suspected or confirmed, levofloxacin should be stopped immediately and appropriate treatment initiated without delay. Anti-peristaltic medicinal products are contraindicated in this clinical situation.

Patients predisposed to seizures

Quinolones may lower the seizure threshold and may trigger seizures. Levofloxacin is contraindicated in patients with a history of epilepsy and, as with other quinolones, should be used with extreme caution in patients predisposed to seizures, or concomitant treatment with active substances that lower the cerebral seizure threshold, such as theophylline. In case of convulsive seizures, treatment with levofloxacin should be discontinued.

Patients with G-6- phosphate dehydrogenase deficiency

Patients with latent or actual defects in glucose-6-phosphate dehydrogenase activity may be prone to haemolytic reactions when treated with quinolone antibacterial agents. Therefore, if levofloxacin has to be used in these patients, potential occurrence of haemolysis should be monitored.

Patients with renal impairment

Since levofloxacin is excreted mainly by the kidneys, the dose of levofloxacin should be adjusted in patients with renal impairment (see **DOSAGE AND ADMINISTRATION**).

Hypersensitivity reactions

Levofloxacin can cause serious, potentially fatal hypersensitivity reactions (e.g. angioedema up to anaphylactic shock), occasionally following the initial dose (see **ADVERSE REACTIONS**). Patients should discontinue treatment immediately and contact their physician or an emergency physician, who will initiate appropriate emergency measures.

Prevention of photosensitisation

Photosensitisation has been reported with levofloxacin. It is recommended that patients should not expose themselves unnecessarily to strong sunlight or to artificial UV rays (e.g. sunray lamp, solarium), during treatment and for 48 hours following treatment discontinuation in order to prevent photosensitisation.

Patients treated with Vitamin K antagonists

Due to possible increase in coagulation tests (PT/INR) and/or bleeding in patients treated with levofloxacin in combination with a vitamin K antagonist (e.g. warfarin), coagulation tests should be monitored when these drugs are given concomitantly.

QT interval prolongation

Caution should be taken when using fluoroquinolones, including levofloxacin, in patients with known risk factors for prolongation of the QT interval such as, for example:

- congenital long QT syndrome
- concomitant use of drugs that are known to prolong the QT interval (e.g. Class IA and III antiarrhythmics, tricyclic antidepressants, macrolides, antipsychotics).
- uncorrected electrolyte imbalance (e.g. hypokalaemia, hypomagnesaemia)
- cardiac disease (e.g. heart failure, myocardial infarction, bradycardia)

Elderly patients and women may be more sensitive to QTc-prolonging medications. Therefore, caution should be taken when using fluoroquinolones, including levofloxacin, in these populations.

Peripheral neuropathy

Cases of sensory or sensorimotor polyneuropathy resulting in paraesthesia, hypaesthesia, dysesthesia, or weakness have been reported in patients receiving quinolones and fluoroquinolones. Patients under treatment with levofloxacin should be advised to inform their doctor and pharmacist prior to continuing treatment if symptoms of neuropathy such as pain, burning, tingling, numbness, or weakness develop in order to prevent the development of potentially irreversible condition (see **ADVERSE REACTIONS**).

Hepatobiliary disorders

Cases of hepatic necrosis up to fatal hepatic failure have been reported with levofloxacin, primarily in patients with severe underlying diseases, e.g. sepsis. Patients should be advised to stop treatment and contact their doctor if signs and symptoms of hepatic disease develop such as anorexia, jaundice, dark urine, pruritus or tender abdomen.

Vision disorders

If vision becomes impaired or any effects on the eyes are experienced, an eye specialist should be consulted immediately.

Superinfection

The use of levofloxacin, especially if prolonged, may result in overgrowth of non-susceptible organisms. If superinfection occurs during therapy, appropriate measures should be taken.

Interference with laboratory tests

In patients treated with levofloxacin, determination of opiates in urine may give false-positive results. It may be necessary to confirm positive opiate screens by more specific method.

Levofloxacin may inhibit the growth of *Mycobacterium tuberculosis* and, therefore, may give false-negative results in the bacteriological diagnosis of tuberculosis.

Development of Drug Resistant Bacteria

Prescribing levofloxacin in the absence of a proven or strongly suspected bacterial infection or a prophylactic indication is unlikely to provide benefit to the patient and increases the risk of the development of drug-resistant bacteria.

Aortic aneurysm and dissection

Epidemiologic studies report an increased risk of aortic aneurysm and dissection after intake of fluoroquinolones, particularly in the older population. Therefore, fluoroquinolones should only be used after careful benefit-risk assessment and after consideration of other therapeutic options in patients with positive family history of aneurysm disease, or in patients diagnosed with pre-existing aortic aneurysm and/or aortic dissection, or in presence of other risk factors or conditions predisposing for aortic aneurysm and dissection (e.g. Marfan syndrome, vascular Ehlers-Danlos syndrome, Takayasu arteritis, giant cell arteritis, Behcet's disease, hypertension, known atherosclerosis).

In case of sudden abdominal, chest or back pain, patients should be advised to immediately consult a physician in an emergency department.

Prolonged, disabling and potentially irreversible serious adverse drug reactions

Very rare cases of prolonged (continuing months or years), disabling and potentially irreversible serious adverse drug reactions affecting different, sometimes multiple body systems (musculoskeletal, nervous, psychiatric and senses) have been reported in patients receiving fluoroquinolones irrespective of their age and pre-existing risk factors. Levofloxacin should be discontinued immediately at the first signs or symptoms of any serious adverse reaction and patients should be advised to contact their prescriber for advice.

USE IN SPECIAL POPULATION

Pregnancy

There are no adequate and well-controlled studies reported in pregnant women. Levofloxacin should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.

Lactation

Based on the reported data on other fluoroquinolones and very limited data on levofloxacin, it can be presumed that levofloxacin will be excreted in human milk. Because of the potential for serious adverse reactions from levofloxacin in nursing infants, a decision should be made whether to discontinue nursing or to discontinue the drug, taking into account the importance of the drug to the mother.

Fertility

Levofloxacin caused no impairment of fertility or reproductive performance in rats.

Paediatrics

Safety and effectiveness in paediatric patients and adolescents <18 have not been established. Quinolones, including levofloxacin cause arthropathy and osteochondrosis in juvenile animals of several species.

Geriatrics

Geriatric patients are at increased risk for developing severe tendon disorders including tendon rupture when being treated with a fluoroquinolone such as levofloxacin. This risk is further increased in patients receiving concomitant corticosteroid therapy. Tendinitis or tendon rupture can involve the Achilles, hand, shoulder, or other tendon sites and can occur during or after completion of therapy; cases occurring up to several months after fluoroquinolone treatment have been reported. Caution should be used when prescribing levofloxacin to elderly patients especially those on corticosteroids. Patients should be informed of this potential side effect and advised to discontinue levofloxacin and contact their healthcare provider if any symptoms of tendinitis or tendon rupture occur.

In a reported study, no overall differences in safety or effectiveness were reported between elderly subjects and younger subjects, but greater sensitivity of some older individuals cannot be ruled out. Severe, and sometimes fatal, cases of hepatotoxicity have been reported post-marketing in association with levofloxacin. The majority of fatal hepatotoxicity reports occurred in patients 65 years of age or older and most were not associated with hypersensitivity. Levofloxacin should be discontinued immediately if the patient develops signs and symptoms of hepatitis.

Elderly patients may be more susceptible to drug-associated effects on the QT interval. Therefore, precaution should be taken when using levofloxacin with concomitant drugs that can result in prolongation of the QT interval (e.g., Class IA or Class III antiarrhythmics) or in patients with risk factors for torsade de pointes (e.g., known QT prolongation, uncorrected hypokalemia).

The pharmacokinetic properties of levofloxacin in younger adults and elderly adults do not differ significantly when creatinine clearance is taken into consideration. However, since the drug is known to be substantially excreted by the kidney, the risk of toxic reactions to this drug may be greater in patients with impaired renal function. Because elderly patients are more likely to have decreased renal function, care should be taken in dose selection, and it may be useful to monitor renal function.

Renal Impairment

Clearance of levofloxacin is substantially reduced and plasma elimination half-life is substantially prolonged in patients with impaired renal function (creatinine clearance <

50 mL/min), requiring dosage adjustment in such patients to avoid accumulation. Neither haemodialysis nor continuous ambulatory peritoneal dialysis (CAPD) is effective in removal of levofloxacin from the body, indicating that supplemental doses of levofloxacin are not required following haemodialysis or CAPD.

Hepatic Impairment

Pharmacokinetic studies in hepatically impaired patients have not been reported. Due to the limited extent of levofloxacin metabolism, the pharmacokinetics of levofloxacin are not expected to be affected by hepatic impairment.

DRUG INTERACTIONS

Effect of other medicinal products on levofloxacin

Chelation Agents: Antacids, Sucralfate, Metal Cations, Multivitamins

While the chelation by divalent cations is less marked than with other fluoroquinolones, concurrent administration of levofloxacin tablets with antacids containing magnesium, or aluminum, as well as sucralfate, metal cations such as iron, and multivitamin preparations with zinc may interfere with the gastrointestinal absorption of levofloxacin, resulting in systemic levels considerably lower than desired. Tablets with antacids containing magnesium, aluminum, as well as sucralfate, metal cations such as iron, and multivitamin preparations with zinc or didanosine may substantially interfere with the gastrointestinal absorption of levofloxacin, resulting in systemic levels considerably lower than desired. These agents should be taken at least two hours before or two hours after oral levofloxacin administration.

Theophylline, fenbufen or similar non-steroidal anti-inflammatory drugs

No pharmacokinetic interactions of levofloxacin were reported with theophylline. However, a pronounced lowering of the cerebral seizure threshold may occur when quinolones are given concurrently with theophylline, non-steroidal anti-inflammatory drugs, or other agents which lower the seizure threshold.

Levofloxacin concentrations were about 13% higher in the presence of fenbufen than when administered alone.

Probenecid and cimetidine

Probenecid and cimetidine had a statistically significant effect on the elimination of levofloxacin. The renal clearance of levofloxacin was reduced by cimetidine (24%) and probenecid (34%). This is because both drugs are capable of blocking the renal tubular secretion of levofloxacin. However, at the tested doses in the study, the statistically significant kinetic differences are unlikely to be of clinical relevance.

Caution should be exercised when levofloxacin is co-administered with drugs that affect the tubular renal secretion such as probenecid and cimetidine, especially in renally impaired patients.

Other relevant information

It has been reported that, pharmacokinetics of levofloxacin are not affected to any clinically relevant extent when levofloxacin was administered together with the following drugs: calcium carbonate, digoxin, glibenclamide, ranitidine.

Effect of levofloxacin on other medicinal products

Ciclosporin

The half-life of ciclosporin was increased by 33% when co-administered with levofloxacin.

Vitamin K antagonists

Increased coagulation tests (PT/INR) and/or bleeding, which may be severe, have been reported in patients treated with levofloxacin in combination with a vitamin K antagonist (e.g. warfarin). Coagulation tests, therefore, should be monitored in patients treated with vitamin K antagonists.

Drugs known to prolong QT interval

Levofloxacin, like other fluoroquinolones, should be used with caution in patients receiving drugs known to prolong the QT interval (e.g. Class IA and III antiarrhythmics, tricyclic antidepressants, macrolides, antipsychotics).

Antidiabetic Agents

Disturbances of blood glucose, including hyperglycemia and hypoglycemia, have been reported in patients treated concomitantly with fluoroquinolones and an antidiabetic agent. Therefore, careful monitoring of blood glucose is recommended when these agents are co-administered.

Other relevant information

In a pharmacokinetic interaction study, levofloxacin did not affect the pharmacokinetics of theophylline (which is a probe substrate for CYP1A2), indicating that levofloxacin is not a CYP1A2 inhibitor.

Other forms of interactions

Food

There is no clinically relevant interaction with food. Levofloxacin tablets may therefore be administered regardless of food intake.

ADVERSE REACTIONS

The information given below is based on data from the reported clinical studies and on extensive post marketing experience.

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

System organ class	Common	Uncommon	Rare	Not known (cannot be estimated from available data)
Infections and infestations		Fungal infection including Candida infection Pathogen resistance		
Blood and lymphatic system disorders		Leukopenia Eosinophilia	Thrombocytopenia Neutropenia	Pancytopenia Agranulocytosis Haemolytic anaemia
Immune system disorders			Angioedema Hypersensitivity	Anaphylactic shock ^a Anaphylactoid shock ^a
Metabolism and nutrition disorders		Anorexia	Hypoglycaemia particularly in diabetic patients	Hyperglycaemia Hypoglycaemic coma)
Psychiatric disorders*	Insomnia	Anxiety Confusional state Nervousness	Psychotic reactions (with e.g. hallucination, paranoia) Depression Agitation Abnormal dreams Nightmares	Psychotic disorders with self-endangering behaviour including suicidal ideation or suicide attempt
Nervous system disorders*	Headache Dizziness	Somnolence Tremor Dysgeusia	Convulsion Paraesthesia	Peripheral sensory neuropathy Peripheral sensory motor neuropathy Parosmia including anosmia Dyskinesia Extrapyramidal disorder Ageusia Syncope

System organ class	Common	Uncommon	Rare	Not known (cannot be estimated from available data)
				Benign intracranial hypertension
Eye disorders*			Visual disturbances such as blurred vision	Transient vision loss
Ear and Labyrinth disorders*		Vertigo	Tinnitus	Hearing loss Hearing impaired
Cardiac disorders			Tachycardia Palpitation	Ventricular tachycardia, which may result in cardiac arrest Ventricular arrhythmia, and torsade de pointes (reported predominantly in patients with risk factors of QT prolongation), Electrocardiogram QT prolonged
Vascular disorders	<i>Applies to iv form only:</i> Phlebitis		Hypotension	
Respiratory, thoracic and mediastinal disorders		Dyspnoea		Bronchospasm, Pneumonitis allergic
Gastrointestinal disorders	Diarrhoea Vomiting Nausea	Abdominal pain Dyspepsia Flatulence Constipation		Diarrhoea – haemorrhagic which in very rare cases may be indicative of enterocolitis, including pseudomembranous colitis Pancreatitis
Hepatobiliary disorders	Hepatic enzyme increased (ALT/AST, alkaline	Blood bilirubin increased		Jaundice and severe liver injury, including cases with fatal acute liver failure, primarily in patients

System organ class	Common	Uncommon	Rare	Not known (cannot be estimated from available data)
	phosphatase, GGT)			with severe underlying diseases Hepatitis
Skin and subcutaneous tissue disorders ^b		Rash Pruritus Urticaria Hyperhidrosis		Toxic epidermal necrolysis Stevens-Johnson syndrome Erythema multiforme Photosensitivity reaction Leukocytoclastic vasculitis Stomatitis
Musculoskeletal and connective tissue disorders*		Arthralgia Myalgia	Tendon disorder including tendinitis (e.g. Achilles tendon) Muscular weakness which	Rhabdomyolysis Tendon rupture (e.g. Achilles tendon) Ligament rupture Muscle rupture
			may be of importance in patients with myasthenia gravis	Arthritis
Renal and urinary disorders		Blood creatinine increased	Renal failure acute (e.g. due to interstitial nephritis)	
General disorders and administration site conditions*	<u>Applies to iv form only:</u> Infusion site reaction (pain, reddening)	Asthenia	Pyrexia	Pain (including pain in back, chest, and extremities)

^a Anaphylactic and anaphylactoid reactions may sometimes occur even after the first dose

^b Mucocutaneous reactions may sometimes occur even after the first dose

*Very rare cases of prolonged (up to months or years), disabling and potentially irreversible serious drug reactions affecting several, sometimes multiple, system organ classes and senses (including reactions such as tendonitis, tendon rupture, arthralgia, pain in extremities, gait disturbance, neuropathies associated with paraesthesia, depression, fatigue, memory impairment, sleep disorders, and impairment of hearing, vision, taste and smell) have been reported in association with the use of fluoroquinolones in some cases irrespective of pre-existing risk factors (see **WARNINGS AND PRECAUTIONS**).

Other undesirable effects which have been associated with fluoroquinolone administration include:

- attacks of porphyria in patients with porphyria.

Other common side effect reported with levofloxacin during reported clinical trials:

Reproductive system and breast disorder: vaginitis

Other less common side effect reported with levofloxacin during reported clinical trials:

Metabolism and Nutrition Disorders: Hyperkalaemia, aggravated diabetes mellitus, dehydration, hypokalaemia, increased LDH and decreased weight

Psychiatric Disorders: Delirium, emotional lability, impotence, manic reaction, mental deficiency, withdrawal syndrome

Nervous System Disorders: Hypertonia, hyperkinesias, abnormal gait, coma, speech disorder, stupor, involuntary muscle contraction

Respiratory, Thoracic and Mediastinal Disorders: Epistaxis

Cardiac Disorders: cardiac arrest, ventricular arrhythmia, hypertension, bradycardia, heart block, supraventricular tachycardia, ventricular fibrillation, angina pectoris, coronary thrombosis, myocardial infarction

Gastrointestinal Disorders: Gastritis, esophagitis gastroenteritis, glossitis, GI haemorrhage, tongue oedema

Renal and Urinary Disorders: Abnormal renal function

Hepatobiliary disorder: Cholethiasis, hepatic coma

Musculoskeletal system disorders: Synovitis, osteomyelitis,

In reported study, ophthalmologic abnormalities, including cataracts and multiple punctate lenticular opacities, have been reported in patients undergoing treatment with quinolones, including levofloxacin. The relationship of the drugs to these events is not presently established.

Other side effect reported with levofloxacin during post-approval use of levofloxacin:

Blood and Lymphatic System Disorders: Aplastic anaemia

Immune System Disorders: Serum sickness

Nervous System Disorders: Exacerbation of myasthenia gravis, isolated reports of encephalopathy, abnormal electroencephalogram (EEG), dysphonia, pseudotumor cerebri

Eye Disorders: Diplopia, visual acuity reduced, vision blurred scotoma

General Disorders and Administration Site Conditions: Multi-organ failure

Investigations: Prothrombin time prolonged, international normalized ratio prolonged, muscle enzymes increased

OVERDOSAGE

According to toxicity studies in animals or clinical pharmacology studies reported with supra-therapeutic doses, the most important signs to be expected following acute overdosage of levofloxacin tablets are central nervous system symptoms such as confusion, dizziness, impairment of consciousness, and convulsive seizures, increases in QT interval as well as gastro-intestinal reactions such as nausea and mucosal erosions.

CNS effects including confusional state, convulsion, hallucination, and tremor have been observed in post marketing experience.

In the event of overdose, symptomatic treatment should be implemented. ECG monitoring should be undertaken, because of the possibility of QT interval prolongation. Antacids may be used for protection of gastric mucosa. Haemodialysis, including peritoneal dialysis and CAPD, are not effective in removing levofloxacin from the body. No specific antidote exists.

STORAGE

Store below 30°C, protected from moisture

PACKING

Blister of 10 x 10's

KEEP ALL MEDICINES OUT OF THE REACH OF CHILDREN

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Product Registration Holder / Manufactured by:

RANBAXY (MALAYSIA) SDN BHD.
Lot 23, Bakar Arang Industrial Estate,
08000 Sungai Petani, Kedah,
Malaysia